

PGow Engineering Technical Report

Comprehensive Summary of Today's Technical Updates & Backend API Specifications

Date: September 2, 2026 | Target System: PGow Backend Services | Author: Antigravity Assistant

1. Executive Overview of Today's Deliverables

Today's development session accomplished major enhancements across **frontend UI/UX**, **KYC document processing & verification**, **accommodation room/bed layout management**, and **codebase design system architectural standardization**. To support these user-facing capabilities, the backend services require specific API endpoints, database schema updates, and state transition workflows.

Module	Frontend Deliverable	Backend Requirement
KYC Verification	Real image enforcement, empty placeholder frames, solid Green/Red action buttons, re-submission workflow for rejected residents.	KYC verification endpoint, rejection reason storage, status state machine (NOT_SUBMITTED → PENDING → VERIFIED/REJECTED).
Bed & Room Layout	Redesigned Bed Visualizer screen with Property Overview, Room Types Filter, Availability Snapshot, and High-Detail Room View Modal.	Layout hierarchy API (`floors` → `rooms` → `beds`), bed allocation/vacate endpoints, and room management APIs.
Architecture & Code Quality	Comprehensive architectural analysis report, full purge of 30+ legacy color aliases in `colors.ts`, standardized LUNA tokens across 22 files.	Clean DTO mapping alignment and strict TypeScript contract compatibility.

2. KYC Document Review & Resident Re-submission Workflow

Context & User Rule: KYC document images were previously unclear or failing to render. The frontend has been updated so that **ONLY actual uploaded document images** (`id\_photo\_url`, `selfie\_url`) are displayed (no fake placeholder photos). When no document photo exists, an explicit dashed placeholder is shown.

Verification Rule: Solid Emerald Green (#10B981) for Verify/Approve and solid Crimson Red (#EF4444) for Reject. When an owner rejects a resident's KYC, the resident is informed of the specific rejection reason and is allowed to re-submit their KYC documents from the beginning.

A. Database Schema Additions (guests table)

```
ALTER TABLE guests ADD COLUMN IF NOT EXISTS kyc_status VARCHAR(30) DEFAULT 'NOT_SUBMITTED'; --
Valid values: 'NOT_SUBMITTED', 'PENDING', 'VERIFIED', 'REJECTED' ALTER TABLE guests ADD COLUMN IF
NOT EXISTS kyc_reject_reason TEXT NULL; ALTER TABLE guests ADD COLUMN IF NOT EXISTS id_type
VARCHAR(50) NULL; ALTER TABLE guests ADD COLUMN IF NOT EXISTS id_number VARCHAR(100) NULL; ALTER
TABLE guests ADD COLUMN IF NOT EXISTS id_photo_url TEXT NULL; ALTER TABLE guests ADD COLUMN IF
NOT EXISTS selfie_url TEXT NULL;
```

B. API Endpoints Specification

1. Resident Submit / Re-submit KYC: POST /v1/kyc

```
// Headers: Authorization: Bearer { "id_type": "AADHAAR", // AADHAAR, PASSPORT, DRIVING_LICENSE,
VOTER_ID "id_number": "1234-5678-9012", "id_photo_url":
"https://storage.pgow.in/kyc/id_123.jpg", "profile_photo_url":
"https://storage.pgow.in/kyc/selfie_123.jpg" } // State Action: Sets kyc_status = 'PENDING',
kyc_reject_reason = null.
```

2. Owner / Manager Verify or Reject KYC: POST /v1/kyc/verify

```
// Headers: Authorization: Bearer { "guest_id": "guest_uuid_123", "approve": false,
"reject_reason": "Photo blurry. Please upload clear Aadhaar front view." } // Response 200 OK: {
"status": "success", "kyc_status": "REJECTED", "message": "KYC rejected. Resident notified to
re-submit." }
```

3. Room & Bed Layout Management System

Context & User Rule: Redesigned the PGow Owner/Manager 'Bed Layout' screen into an intuitive, visually engaging room-management experience while preserving all backend data hooks and allocation logic.

Tapping **Vacant** filters for vacant beds; tapping **Occupied** filters for occupied beds. Filter chips allow browsing by sharing type (Single, 2 Sharing, 3 Sharing, 4 Sharing). Room Cards open a High-Detail Room Modal with 2D Floorplan previews, bed assignment, resident transfer, and history controls.

A. Relational Data Model (Floors → Rooms → Beds)

```
CREATE TABLE floors ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), pg_id UUID NOT NULL
REFERENCES properties(id), floor_number INT NOT NULL ); CREATE TABLE rooms ( id UUID PRIMARY KEY
DEFAULT gen_random_uuid(), pg_id UUID NOT NULL REFERENCES properties(id), floor_id UUID NOT NULL
REFERENCES floors(id), room_number VARCHAR(20) NOT NULL, sharing_type VARCHAR(20) NOT NULL, --
'SINGLE', '2_SHARING', '3_SHARING', '4_SHARING' base_rent DECIMAL(10,2) NOT NULL ); CREATE TABLE
beds ( id UUID PRIMARY KEY DEFAULT gen_random_uuid(), room_id UUID NOT NULL REFERENCES rooms(id),
bed_number VARCHAR(10) NOT NULL, -- 'Bed A', 'Bed B' is_occupied BOOLEAN DEFAULT FALSE,
tenant_membership_id UUID NULL REFERENCES memberships(id), check_in_date DATE NULL );
```

B. API Endpoint: Fetch Property Layout

Endpoint: GET /v1/properties/:pgId/layout

```
{ "pg_id": "pg_999", "total_floors": 3, "total_rooms": 12, "total_beds": 36, "occupied_beds":
24, "vacant_beds": 12, "occupancy_rate": 66.6, "floors": [ { "floor_number": 1, "rooms": [ {
"id": "room_101", "room_number": "101", "sharing_type": "2_SHARING", "base_rent": 8500, "beds":
[ { "id": "bed_101_a", "bed_number": "Bed A", "is_occupied": true, "tenant_name": "Rahul Sharma"
}, { "id": "bed_101_b", "bed_number": "Bed B", "is_occupied": false, "tenant_name": null } ] } ] }
```

C. Bed Allocation API Endpoints

1. Assign Bed: POST /v1/properties/:pgId/beds/:bedId/assign Payload: { "tenant_membership_id": "mem_789" } 2. Vacate Bed: POST /v1/properties/:pgId/beds/:bedId/vacate 3. Create Room: POST /v1/properties/:pgId/rooms Payload: { "floor_number": 1, "room_number": "105", "sharing_type": "2_SHARING", "base_rent": 8500 }

4. Codebase Architecture & LUNA Token Cleanup

Completed a comprehensive code quality & architectural overhaul of the **PGow-Frontend** codebase:

- Generated detailed architecture analysis artifact (codebase_architectural_analysis.md).
- Stripped **30+ legacy color aliases** (e.g. CyberGreen, LuxuryPureBlack, SlateMutedText, res*) from colors.ts.
- Reduced colors.ts from 208 lines to 144 clean lines.
- Replaced legacy tokens across **22 files** with strict semantic LUNA tokens (Colors.primary, Colors.canvas, Colors.textPrimary, Colors.borderSubtle).
- Verified **0 legacy token references** remaining in the entire src/ directory.

5. Backend Engineering Action Item Checklist

#	Task Item	Priority	Target Table / Route
1	Add kyc_status, kyc_reject_reason columns	HIGH	guests table
2	Implement POST /v1/kyc endpoint	HIGH	/v1/kyc
3	Implement POST /v1/kyc/verify with reject reason	HIGH	/v1/kyc/verify
4	Build nested Property Layout endpoint	HIGH	/v1/properties/:pgId/layout
5	Implement Bed Assign & Vacate endpoints	MEDIUM	/v1/properties/:pgId/beds/*
6	Implement Room Creation & Sharing update APIs	MEDIUM	/v1/properties/:pgId/rooms